Example: $h_{a,b}(m) = a \cdot m + b \bmod p$

Let p be a large prime ($\geq |b \cup T_S|$)

$(a, b) \in \mathbb{Z}_p^2 = K$

$m \in \mathbb{Z}_p = \mathcal{M} = \mathcal{T}$.

LEMMA Above $\mathcal{H}$ is PAIRWISE INDEPENDENT.

Proof. Fix any $m, m' \in \mathbb{Z}_p$, $m \neq m'$, $\tau, \tau' \in \mathbb{Z}_p$.

$$\Pr_{a,b}\left[h_{a,b}(m) = \tau \wedge h_{a,b}(m') = \tau' \right]$$

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$$\Pr_{a,b}\left[\begin{pmatrix} m & 1 \\ m' & 1 \end{pmatrix}\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} z \\ z' \end{pmatrix}\right]$$

$$\Pr_{a,b}\left[\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} m & 1 \\ m' & 1 \end{pmatrix}^{-1}\begin{pmatrix} z \\ z' \end{pmatrix}\right]$$

$\Leftarrow$ The inverse exists

$\Leftrightarrow$ $m \neq m'$

$= 1/p^2$ ∎

THM Every PAIRWISE INDEPENDENT $\mathcal{H}$ defines an ε-sec. secure MAC with $\varepsilon = 1/|\mathcal{T}|$

$$\mathcal{H} = \{ h_K : \mathcal{M} \to \mathcal{T} \}$$

Proof. By Bayes, $\forall\ m, m', \tau, \tau'$ $(m \neq m')$

$$\Pr[\tau' = h_K(m') \mid \tau = h_K(m)] =$$

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$$\frac{\Pr[\tau' = h_k(m') \wedge \tau = h_k(m)]}{\Pr[h_k(m) = \tau]} =$$

$$\frac{1/|\mathcal{T}|^2}{1/|\mathcal{T}|} = \frac{1}{|\mathcal{T}|}$$

□

Bel news:

Thm Any t-time $2^{-\lambda}$-secure MAC has a key of size $(t+1)\cdot\lambda$.

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RANDOMNESS EXTRACTION

How do Alice & Bob get random bits?

General to CRYPTO:

- Keys must be UNIFORM.
- Randomness also later needed for security (w.r. CPA Security).

You can think about this as a "2-step process":

1) Randomness extraction: Starting

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from IMPERFECT source of randomness, generate "pure" RANDOMNESS.

2) Expand what you extracted as needed deterministically. (PSEUDORANDOMNESS).

3) Plus some way to refresh the RANDOMNESS that you can again expand.

Here is the question: We have a source X and we would like to design Ext s.t. Ext(X) is UNIFORM.

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Impossible in general (e.g. X = constant).

X should be unpredictable to some extent.

Def (min-entropy) The min-entropy of X

$$H_\infty(X) \doteq -\log \max_x \Pr[X = x]$$

Intuition: The more the min-entropy the better. For example:

Let $X = 0^n$; $H_\infty(X) = 0$

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Let $X \equiv U_n \quad (X \leftarrow \{0,1\}^n)$

$$H_\infty(X) = n$$

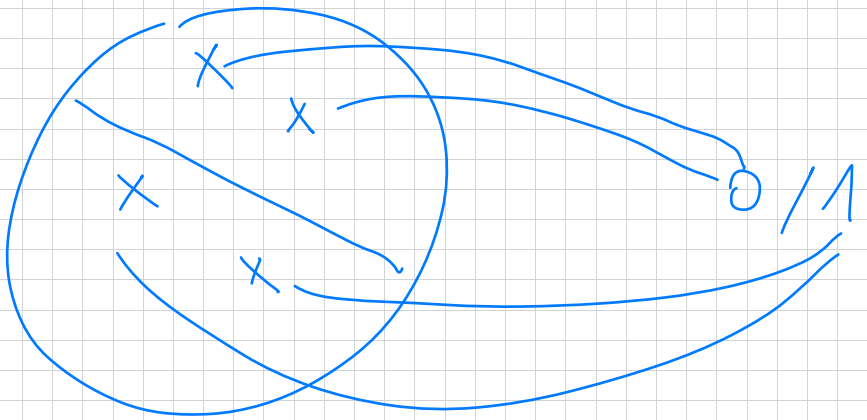
Typically: One knows $H_\infty(X) \geq k$ for some $k < n$ and we'd like to extract as much randomness as possible.

THM. There is no such $Ext : \{0,1\}^n \to \{0,1\}$ even if $k = n-1$.

Proof. Take any $Ext : \{0,1\}^n \to \{0,1\}$.

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Let $b \in \{0,1\}$ s.t. b maximizes $|Ext^{-1}(b)|$



$$\Rightarrow |Ext^{-1}(b)| \geq 2^{n-1}$$

Let X_{BAD} be uniform over $Ext^{-1}(b)$.

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$H_\infty(X_{BAD}) \geq n-1$ but

what is $Ext(X_{BAD}) = b$ constant!

TM

Take-away: No Ext can fool all min-entropy sources. Approaches:

- Assume multiple independent sources. 2-SOURCE EXTRACTORS.
- Use crypto tools (e.g. HASH

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FUNCTIONS). But requires strong assumptions on these functions.

- SEEDED EXTRACTION. We cover this.

Idea: Family of extractors

$$Ext : \underbrace{\{0,1\}^{d}}_{\text{SEED}} \times \underbrace{\{0,1\}^{n}}_{\text{SOURCE}} \to \underbrace{\{0,1\}^{\ell}}_{\text{OUTPUT}}$$

The seed is going to be RANDOM.

Why not cheating:

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- Seed is short catalyst; output can be longer.

- Seed is going to be PUBLIC and REUSABLE. (The Adv knows it!)

DEF $Ext: \{0,1\}^d \times \{0,1\}^n \to \{0,1\}^\ell$ is a (k, ε) SEEDED-extractor if $\forall X$ s.t. $H_\infty(X) \geq k$ we have:

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$(S, Ext(S, X)) \tilde{\approx}_\varepsilon (S, U_\ell) \; (\equiv U_{d+\ell})$

where $S \leftarrow U_d$

$\forall$ UNBOUNDED A can distinguish $(S, Ext(S, X))$ from $U_{d+\ell}$ w.p. $\leq \varepsilon$.

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$$X \approx_\varepsilon Y \iff SD(X;Y) \leq \varepsilon$$

$$SD(X;Y) = \frac{1}{2} \sum_z \left| Pr[X=z] + \right.$$
$$\left. - Pr[Y=z] \right|$$

THM (LEFTOVER HASH LEMMA) Let

$$\mathcal{H} = \{ h_s : \{0,1\}^n \to \{0,1\}^\ell \}_{s \in \{0,1\}^d}$$

be PAIRWISE INDEPENDENT.

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That $Ext(s,x) = h_s(x)$ is a (k, ε) – seeded extractor for

$$k \geq l + \underbrace{2\log(1/\varepsilon) - 2}_{\text{MIN-ENTROPY GAP}}$$

The proof will use the following lemma:

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LEMMA Let Y be a RV over $\mathcal{Y}$.

Assume:

$$Col(Y) = \sum_{y \in \mathcal{Y}} Pr[Y = y]^2$$

$$\text{UNIFORM over } \mathcal{Y} \quad \leq \frac{1}{|\mathcal{Y}|} (1 + 4\varepsilon^2)$$

$$\Rightarrow SD(Y; U) \leq \varepsilon.$$

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Proof (THM). Set $Y = (S, h(S,X))$

$$Col(Y) = Pr[Y = Y']$$

$$= Pr[S = S' \wedge h(S,X) = h(S',X')]$$

$$= Pr[S = S' \wedge h(S,X) = h(S,X')]$$

$$= \underbrace{Pr[S = S']}_{Col(S)} \cdot Pr[h(S,X) = h(S,X')]$$

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$$= 2^{-\ell} \cdot \Big(\Pr[X = X'] \cdot 1 \; + \; \Pr[h(S,X) = h(S,X') \wedge X \neq X'] \Big)$$

$H_\infty(X) \geq k$

$$\leq 2^{-\ell} \Big(2^{-k} + \Pr[h(S,X) = h(S,X') \wedge X \neq X'] \Big)$$

$$\leq 2^{-\ell} \left(2^{-k} + 2^{-\ell} \right)$$

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$$= \frac{1}{2^{d+l}} \left(2^{l-k} + 1 \right)$$

$$\hookrightarrow l - k \leq 2 - 2\log(1/\varepsilon)$$

$$\leq \frac{1}{|\mathcal{Y}|} \left(2^{2-2\log(1/\varepsilon)} + 1 \right)$$

$$= 1/|\mathcal{Y}| \left(1 + 4\varepsilon^2 \right)$$

$$\Rightarrow SD(Y; U) \leq \varepsilon \qquad \blacksquare$$

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